

MAD 2 Project Report

1. Student Details

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College: VIT Pune & IIT Madras BS in Data Science

2. Project Details

Project Title: Trekking Management System

Problem Statement:

Develop a trekking management web application supporting Admin, Trek Staff and Users with authentication, trek management, bookings, dashboard, CSV export and background jobs.

Approach:

Implemented using Flask, Vue.js, SQLAlchemy, SQLite, JWT, Redis and Celery. The application provides role-based access, REST APIs, booking workflow, profile management, CSV export, daily reminder emails and monthly HTML reports sent to the administrator.

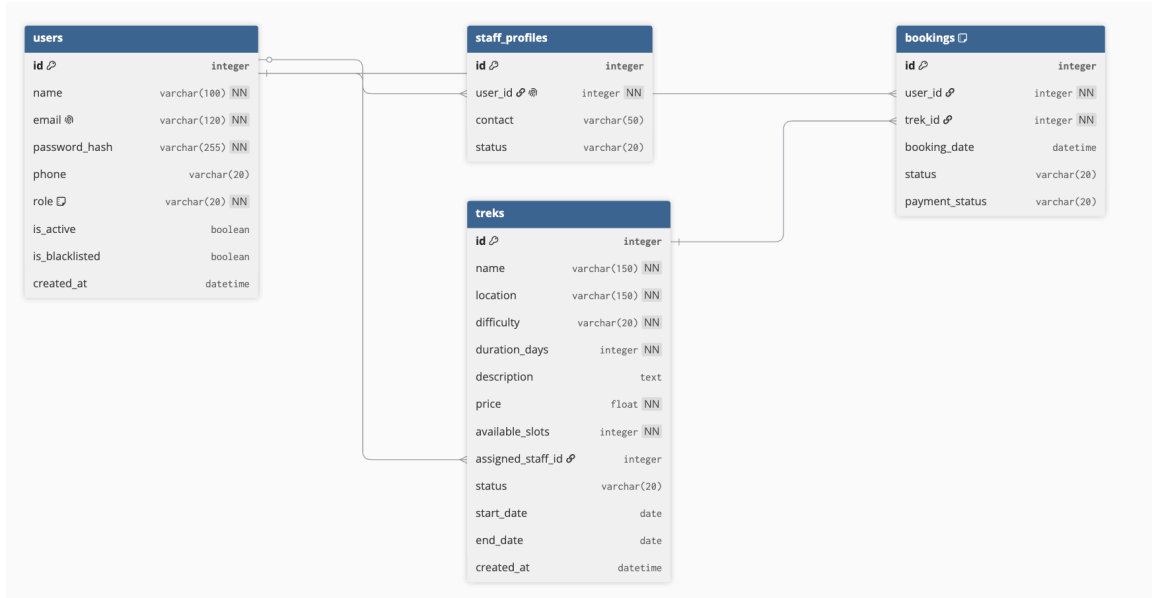
3. AI / LLM Declaration

AI tools (primarily ChatGPT) were used only as a learning and development aid. Assistance included understanding Flask, Vue.js, Celery, Redis, JWT authentication, SQLAlchemy concepts, debugging runtime errors, fixing integration issues, improving code quality, and refining documentation. The overall system design, database schema, implementation, testing, integration and final submission were completed by me.

4. Frameworks and Libraries Used

Framework	Purpose
Flask	Backend
Vue.js	Frontend
SQLAlchemy	ORM
SQLite	Database
Flask-JWT-Extended	JWT Authentication
Redis	Broker
Celery	Background Jobs
Bootstrap	UI
Axios	API Calls

5. ER Diagram



6. API Resource Endpoints

Endpoint	Method	Description
/api/auth/register	POST	Register
/api/auth/login	POST	Login
/api/auth/me	GET	Profile
/api/treks	GET	Treks
/api/bookings	POST	Book trek
/api/bookings/history	GET	History
/api/bookings/export	GET	CSV Export
/api/admin/*	Various	Admin APIs
/api/staff/*	Various	Staff APIs

7. Video Presentation

Google Drive Link:

https://drive.google.com/file/d/1_M93wY9IHPtHBie3YatXkMQ7Te4REgum/view?usp=sharing